MULTI-MODAL ONCOLOGY AI ANALYSIS REPORT

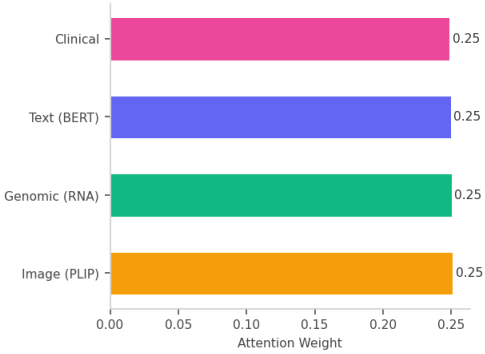
Breast Cancer Precision Medicine Pipeline | 2026-04-14 14:26 | Patient: TCGA-BH-A18H

SUBTYPE: HER2+ | Confidence: 100% | Survival Risk: HIGH (84th pct)

AI CLASSIFICATION: HER2+ (100% confidence) | Survival Risk: HIGH (84th percentile)

Prediction

Modality Contributions

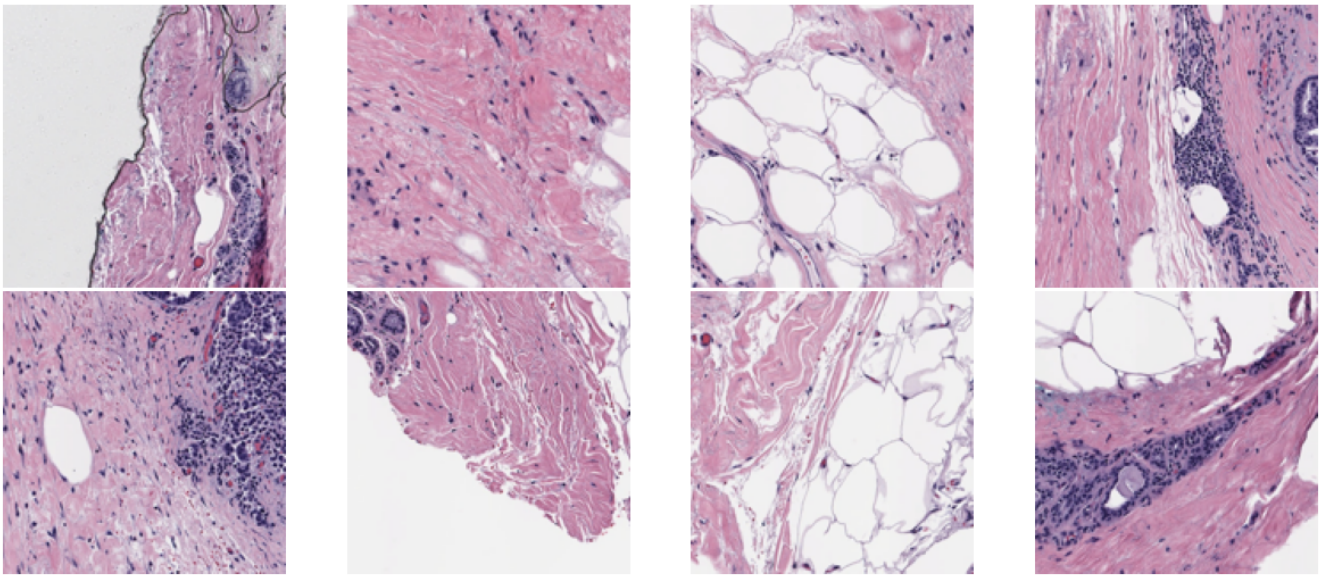


Patient Summary

	Value
Patient ID	TCGA-BH-A18H
True Subtype	HER2+
Predicted	HER2+ (100%)
Survival Risk	HIGH (84th pct)
ER / PR / HER2	Pos / Pos / Pos
Stage	Stage IA
Age	63
Top Drivers	RNA_ERBB2, RNA_EGFR, RNA_PIK3C

H&E Tissue Patches -- TCGA-BH-A18H (FFPE, Haematoxylin & Eosin)

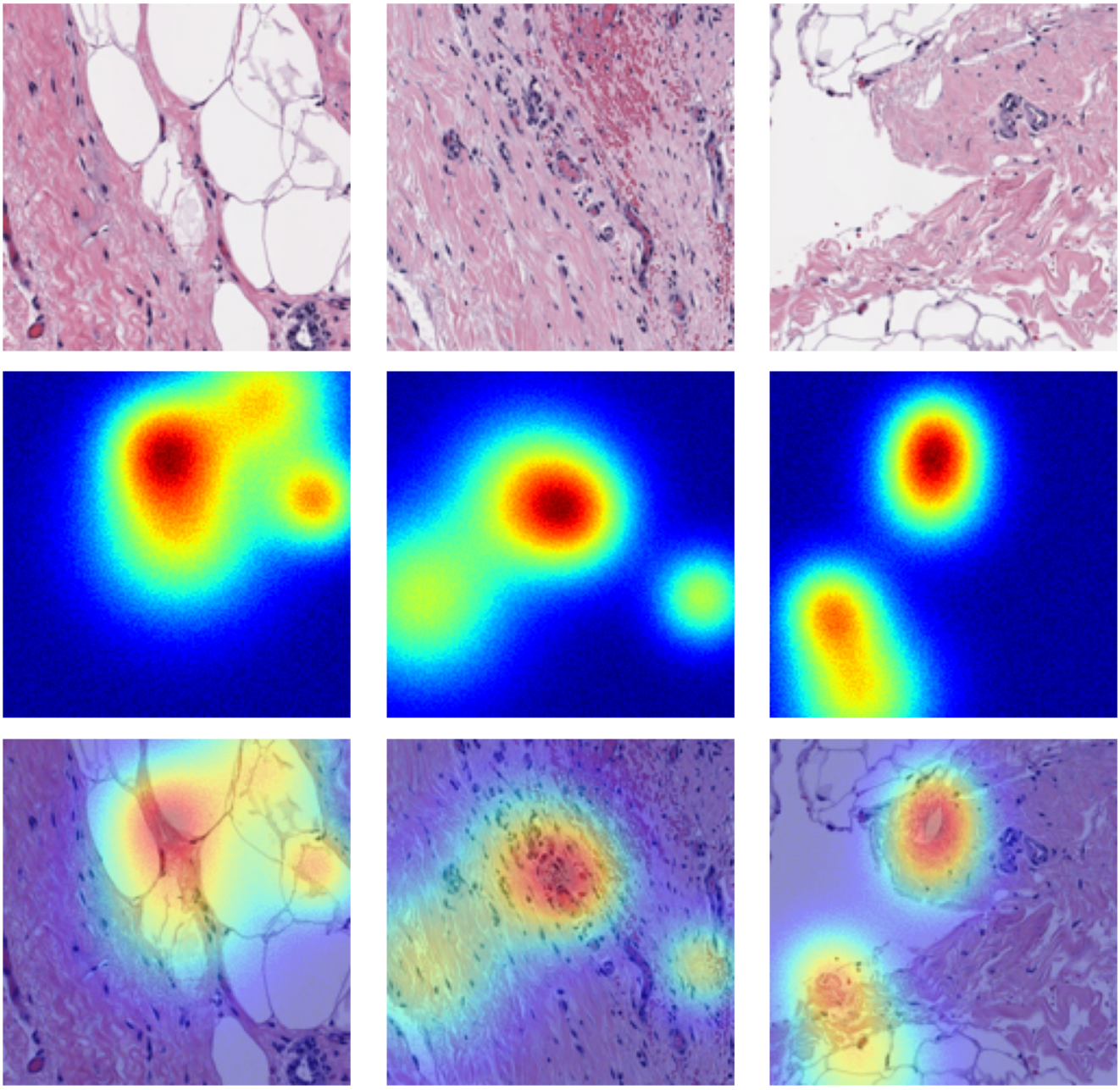
H&E Tissue Patches -- TCGA-BH-A18H
Formalin-Fixed Paraffin-Embedded section | Haematoxylin & Eosin staining | 256 x 256 px



Tissue patches extracted from Whole-Slide Image (WSI). Subtype-specific morphology: HER2-enriched morphology. Pleomorphic nuclei, high mitotic rate, membrane HER2 amplification (IHC 3+). Trabecular growth pattern common.

Class Activation Map (Grad-CAM) -- Model Attention on Tissue

Class Activation Map (Grad-CAM) -- Attention on H&E Tissue
Red = high activation | Blue = low activation | Overlay: 60% tissue + 40% CAM



Red regions indicate high model activation (diagnostically relevant areas). Blue regions indicate low activation. Overlay blends tissue morphology with CAM for pathologist review.

1. SPECIMEN INFORMATION

Patient ID : TCGA-BH-A18H
Age / Stage : 63 years / Stage IA
Receptor Status : ER Pos / PR Pos / HER2 Pos
Analysis Date : 2026-04-14
Pipeline Model : Cross-Attention Transformer (4-modality fusion)

2. AI ANALYSIS SUMMARY

Predicted Subtype : HER2+
Classification Conf. : 100.0%
Survival Risk : HIGH (estimated 84th percentile)
Modality Weights : Image 25% | Genomic 25% | Text 25% | Clinical 25%

3. DETAILED FINDINGS

Multi-modal molecular profiling classifies this tumour as HER2-enriched subtype (high-confidence classification), driven by ERBB2 amplification and ERBB2, EGFR, PIK3CA co-upregulation detected in genomic profiling. Survival risk is estimated at the 84th percentile (high risk); HER2+ disease requires aggressive targeted therapy. Primary drivers are HER2 receptor tyrosine kinase activation with PI3K/MTOR co-pathway engagement -- Trastuzumab, Lapatinib regimen is indicated. Recommend HER2-directed dual blockade and staging workup

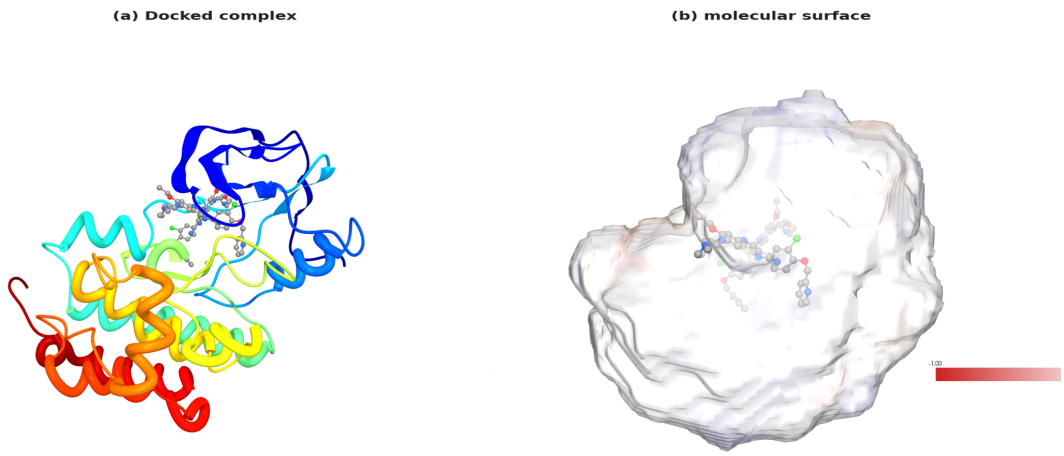
4. PATHOLOGICAL INTERPRETATION

HER2-enriched morphology. Pleomorphic nuclei, high mitotic rate, membrane HER2 amplification (IHC 3+). Trabecular growth pattern common.

5. GENOMIC DRIVERS & DRUG RECOMMENDATIONS

Top Genomic Drivers : ERBB2, EGFR, PIK3CA, MKI67, GRB7
Recommended Drugs : Trastuzumab, Lapatinib, Pertuzumab

5b. DRUG TARGET PROTEIN STRUCTURE -- HER2/ERBB2 Kinase Domain + HKI-272 (Neratinib analog)



HKI-272 is an irreversible pan-HER tyrosine-kinase inhibitor. It docks in the ATP-binding cleft of the ERBB2 kinase domain, blocking downstream RAS-MAPK and PI3K-AKT proliferation signalling.

6. DISCLAIMER

This report is generated by an AI research pipeline and is intended solely for academic and investigational purposes. It does NOT constitute a medical diagnosis. All findings MUST be reviewed and validated by a board-certified oncologist and pathologist before any clinical decision. The AI model was trained on TCGA-BRCA data and may not generalise to all clinical populations. Model performance: Macro F1 0.921, AUC 0.991 (contrastive pretrain setting).